

AI-Powered Semantic Reconciliation System for Pakistani SMEs

24I-0767, 24I-0899, 24I-0581 | Software Design and Analysis (CX01)

1. Project Overview

We are developing an AI-centric reconciliation engine to automate financial data matching for Pakistani SMEs. The system addresses the "reconciliation crisis" where 88% of retail transactions are digital but reconciliation still remains manual. Our solution uses a RAG model to bridge bank statements and internal ledgers, a task where traditional rule-based systems fail.

2. Motivation

Though many ERP softwares have sped up the process, none leverage LLMs to utilize semantic matching to speed up the process. Pakistani businesses currently face four critical challenges:

- 1. Manual Inefficiency:** Finance teams spend **40-60 hours monthly** transcribing WhatsApp payment screenshots into Excel, resulting in severe audit trail fragmentation.
- 2. Economic Burden:** With an average accountant salary of **PKR 1.67M annually**, employing 2-3 staff for reconciliation drives costs to **PKR 3.3-5M per year**. Processing just 750 monthly transactions consume over **81 hours** of manual labor.
- 3. Regulatory Risk:** FBR's mandatory **Annexure-C** requires real-time NTN/STRN/HS code submission. Manual processes result in annual penalties ranging from **PKR 150K-300K**.

3. Objectives

Primary Goal: Achieve 85%+ automatic matching accuracy, reducing manual reconciliation workload by 80-90%.

Technical and Business Objectives:

- 1. Image Processing:** Process compressed WhatsApp images with 90%+ OCR accuracy.
- 2. AI Optimization:** Fine-tune Alif-1.0-8B to achieve >0.85 confidence on 85% of transactions. Deploy Gemini-3-Pro as the final validation layer before human input
- 3. Payment Integration:** Integrate Raast P2M APIs to capture real-time transaction metadata.
- 4. Regulatory Automation:** Automate FBR Annexure-C submission via the IRIS portal.
- 5. Operational Speed:** Reduce the monthly closing cycle from 15 days to 2-3 days.

4. Expected Outcomes

Quantitative: 85-90% auto-match rate, <10 sec per transaction, errors reduced from 0.5% to <0.1%, 70+ hours monthly labor savings (81→<10 hrs), 100% on-time FBR submissions, <2% false positives.

Qualitative: Complete audit trail, real-time cash visibility, finance staff redeployed to strategic analysis, enhanced audit readiness, scalable infrastructure for 10x growth.

Deliverables: A functional prototype capable of end-to-end processing and a "Mock" Banking API and FBR Portal interface to demonstrate integration logic without requiring live regulatory approval

5. Project Scope

In Scope: OCR extraction (WhatsApp screenshots, PDFs, JPEG/PNG), bilingual text (English/Urdu/Roman Urdu), LLM semantic matching with confidence scoring, CSV/Excel export, review UI. **Out of Scope:** Full ERP/GL functionality, EasyPaisa/JazzCash APIs (Phase 2), invoice generation, AP workflows, fraud detection, on-premise deployment (cloud-only). **Constraints:** Target: PKR 500M-2B revenue firms, 500-1000 monthly txns. 8-week pilot, 40%+ Raast adoption assumed, <PKR 1M budget.

6. Tools and Technologies

Our software will be built on an Enterprise-Modern architecture that prioritizes financial reliability while leveraging localized intelligence to solve the "Semantic Gap" in Pakistani transaction data. At its core, we utilize Java 21 and Spring Boot 3 to provide a strictly typed, ACID-compliant backend. Rather than relying on rigid, error-prone "If/Then" rules, our system employs a Probabilistic AI engine powered by Alif-1.0-8B, a state-of-the-art LLM specifically fine-tuned for the cultural and linguistic nuances of Urdu and Roman Urdu, and Gemini 3 Pro as a foolproof, validation layer. By integrating this intelligence via a RAG (Retrieval-Augmented Generation) pipeline with a PostgreSQL and pgvector hybrid database, this allows us to reconcile the chaotic reality of manual bank memos and fragmented ledger entries into a verified, audit-ready state with a speed and accuracy that traditional accounting systems cannot match.